INCH-POUND

MIL-A-82842(OS)
5 February 1990
SUPERSEDING
WS 21186
8 November 1983
(See 6.8)

MILITARY SPECIFICATION

ADHESIVE, CONDUCTIVE, LOW RESISTIVITY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of low resistivity, electrically conductive, two component adhesive, referred to herein as the adhesive (see 6.1.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

SPECIFICATIONS

FEDERAL

O-T-620 1,1,1 Trichloroethane Technical, Inhibited (Methyl Chloroform)

(Unless otherwise indicated, copies of federal and military specifications standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001).

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 792 Specific Gravity (Relative Density) and Density of Plastics by Displacement

ASTM D 1002 Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal)

ASTM D 1475 Density of Paint, Varnish, Lacquer, and Related Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

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AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677).

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the adhesive shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The adhesive shall consist of an organic resin containing a conductive filler and a fast heat cure catalyst which, when mixed in the recommended ratio, hardens to form a conductive material with low electrical resistance characteristics.

3.3 Physical and electrical properties. Physical and electrical properties of the adhesive shall conform to table I.

3.4 Shelf life. The adhesive shall have a shelf life of 6 months after date of manufacture when stored at 80 degrees Fahrenheit (°F) maximum.

3.4.1 Shelf life extension. The shelf life expiration date of an individual lot may be extended for additional 6 month periods provided the adhesive, upon retest, conforms to the volume resistivity and tensile shear strength requirements of table I.

3.5 Safety

3.5.1 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions

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pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

TABLE I. Physical and electrical properties.

Property	Limits	
	Minimum	Maximum
Base compound properties		
Density, grams/cubic centimeter (g/cc)	3.0	4.0
Mixed compound properties		
Working life, minutes	40	---
Consistency	Paste	
Cured materials properties		
Density, g/cc	3.0	4.0
Tensile shear strength, Pounds per square inch (psi)	600	---
Volume resistivity, ohm-centimeter (cm)	---	0.02

3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.4).

3.6 **Workmanship.** Workmanship shall be such that the adhesive is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1. **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table II. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II.

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of all the adhesive offered for acceptance at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 Selection of sample containers. Sample containers of the adhesive shall be selected at random from each lot in accordance with table III for the inspections specified in 4.6.1 through 4.6.6. Sample material shall be selected as specified in 4.4.1.3.

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TABLE II. First article and quality conformance inspections.

Examination and tests	Requirement	Test Method
Visual examination	3.6 and section 5	4.6.1
Density, base resin test	3.3	4.6.2
Working life and consistency test	3.3	4.6.3
Cured material density test	3.3	4.6.4
Tensile Shear strength test	3.3	4.6.5
Volume resistivity test	3.3	4.6.6

TABLE III. Sampling for inspection.

Number of samples in the lot	Number of samples to be selected for inspection
1 - 3	All
4 - 40	3
41 - 65	4
66 -110	5

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.6.1 through 4.6.6.

4.4.1.4 **Noncompliance of samples.** Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

4.5 **Test conditions.** Unless otherwise specified, all tests shall be conducted at 75 ± 5°F. All test specimens shall be conditioned for 24 hours ± 4 at 75 ± 5°F and 50 ± 10 percent relative humidity prior to testing.

4.6 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

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NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.6.1 Visual examination. All containers shall be visually examined for conformance to 3.5, 3.6 and section 5.

4.6.2 Density, base resin test. The density of the base material shall be determined in accordance with ASTM D 1475.

4.6.3 Working life and consistency test. The working life of the mixed adhesive shall be determined by mixing a 10 ± 2 gram sample on a glass slide. The workability of the sample shall be tested every 5 minutes until the material is no longer spreadable. The sample shall be visually inspected for consistency.

4.6.4 Density, cured compound test. The density of the cured adhesive shall be determined in accordance with ASTM D 792, Method A-1.

4.6.5 Tensile shear strength test. Tensile shear strength shall be determined in accordance with ASTM D 1002 and the following:

a. Prepare test panels using 2024T3 or 6061 aluminum 0.125 ± 0.005 inch thick.

b. Roughen test panel surfaces with 100 grit cloth.

c. Wipe the test panels with a suitable solvent such as O-T-620 methyl chloroform or equivalent.

d. Mix adhesive components in accordance with manufacturer's instructions. Mix thoroughly until uniformly blended.

e. Apply thin coat of adhesive on bonding surface of each panel, and assemble panels in pairs using overlap of 0.50 ± 0.010 inch. Apply a small clamping force not to exceed 1 psi prior to curing of test specimens.

f. Cure test specimens for a minimum of 2 hours at 140 ± 10°F.

g. Use metal shims, 0.125 inch thick, to align specimens in Instron jaws prior to testing.

h. Record individual test specimens results to the nearest psi.

4.6.6 Volume resistivity test. Volume resistivity shall be determined in accordance with the following:

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a. Across one face of a cured epoxy resin block which is 6 ± 1.2 inches long, mill slots 1/8 inch deep by 1/8 inch wide by 6 ± 1.2 inches long.

b. Sand a No. 12 copper wire with No. 100 emery cloth and place one piece of wire, one-half inch long, at each end of the slot so the wire extends 0.25 ± 0.05 inch from end of the slot.

c. Mix the material to be tested in accordance with 4.6.5d.

d. Immediately after mixing, fill the slots with the mixed material.

e. Cure the assembly a minimum of 2 hours at 140 ± 10°F.

4.6.6.1 **Test procedures.** Volume resistivity may be measured by either of the following methods:

a. Measure the resistance by attaching an ohmmeter or Wheatstone Bridge to wires extending from the specimen and calculate the volume resistivity as follows:

$$\text{Volume resistivity} = \frac{\text{Ohms x cross sectional area } (cm^2)}{\text{length (cm)}}$$

b. As shown on figure 1, attach the spring copper contacts to wires extending from the specimen.

(1) Apply voltage (P) to give a voltage drop (V) equal to 10.20 millivolt (mv).

(2) Calculate the volume resistivity as follows:

$$\text{Volume resistivity} = \frac{V \times D}{L \times A}$$

Where: D = cross sectional area of specimen (square cm)

L = distance between metal prods (cm)

NOTE: Prods must not be closer to the ends of specimen than 1 cm.

V = voltage drop in specimen (volts)

P = voltage applied to specimen (volts)

A = current flow through specimen (amperes)

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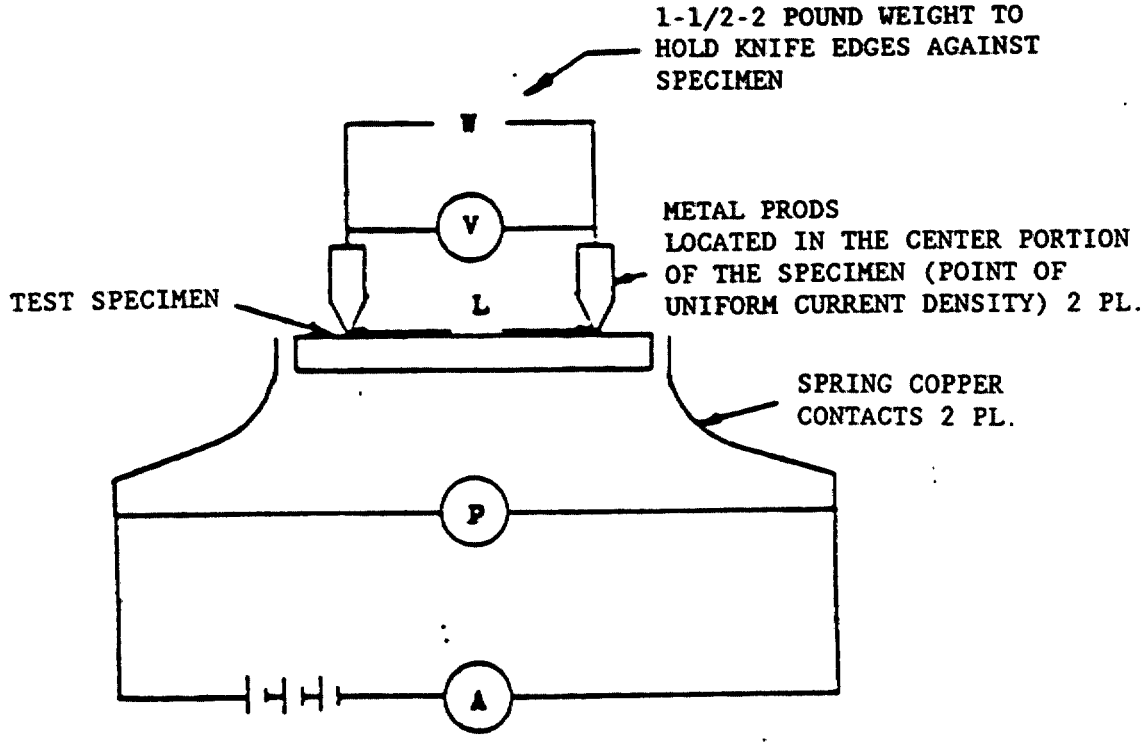


FIGURE 1. Test procedure setup.

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c. Record the volume resistivity to nearest 0.0001 ohm-cm.

4.7 **Inspection of packaging.** The adhesive containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The adhesive shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000 National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), the adhesive shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each adhesive marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ADDITIONAL 6 MONTH PERIODS PROVIDED THE ADHESIVE, UPON RETEST, CONFORMS TO THE VOLUME RESISTIVITY AND TENSILE SHEAR STRENGTH REQUIREMENTS OF TABLE I SPECIFIED IN MIL-A-82842."

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The adhesive specified herein is intended for use as a conductive adhesive in the manufacture of Standard Missile Mk 104 Dual Thrust Rocket Motors.

6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheets (see 3.5.2 and 6.4).

6.3. First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the adhesive sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheet (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Definitions.

6.5.1 Batch. A batch is that amount of adhesive made as one unit in the manufacturing process.

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6.5.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.6 Possible material. The following material has been found to meet the requirements of this specification. This is only given for information and is not restrictive.

ECOBOND SOLDER 56C manufactured by Emerson and Cuming.

6.7 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.8 Supersedure information. MIL-A-82842 incorporates the following engineering change proposal (ECP) and specification change notice (SCN):

ECP	SCN
SSS1026 8/22/84	SCN 1 8/22/84

Preparing Activity:
NAVY-OS
(Project 1340-N154)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82842	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, CONDUCTIVE, LOW RESISTIVITY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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